

Harith Al-Safi

📍 Manchester, UK 📞 +447444585915 @ harith.alsafi@gmail.com 🌐 https://harith.io/

Electronic & Computer Engineer | Software Developer 🐙 github.com/harith-alsafi 🌐 linkedin.com/in/harith-al-safi

📄 About me

An experienced Electronic & Computer Engineer specializing in full-stack development and AI/ML applications, with a proven track record in developing high-performance systems and leading technical teams.

📁 Experience

Fullstack Engineer

BT Group

📅 Sep 2024 - Ongoing

📍 Manchester, UK

- Architected high-performance REST APIs using Java Spring Boot with MongoDB achieving 99.9 % uptime
- Implemented deployment architecture using Pulumi, Kafka and AWS improving response time by 60 %
- Established mentorship program for apprenticeship students, resulting in 90 % pass rate for training assessments
- Led regional Grad network initiatives, organizing monthly tech talks and workshops with 50+ regular attendees

Frontend Developer

Velz Travel

📅 Aug 2023 - Nov 2023 (4 Months)

📍 Remote

- Directed the frontend of a travel web app for a startup built with Next.js, React.js and Typescript
- Developed user interface using Material UI following Figma design guidelines and business logic using Node.js
- Collaborated with developers using Jira and Git whilst working on Java Springboot with PostgreSQL and Docker
- Achieved 93% test coverage through comprehensive unit testing, resulting in 70% reduction in production bugs

Software Engineer

Johnson Controls

📅 July 2022 - July 2023 (12 Months)

📍 London UK

- Administered C \C++ projects while fixing bugs, adding features, refactoring and migrating 90% of codebase to C#
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an analytics web dashboard with React.js for predictive maintenance with dev ops and CI/CD
- Invented a custom scripting language with a compiler for C# app increasing development productivity by 40%
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure
- Managed a team of 10 interns to create a business plan and ended up placing 2nd amongst the whole company

Software Developer Freelance

Paperound

📅 Dec 2021 - Mar 2022 (4 Months)

📍 Leeds UK

- Developed full stack web app using React.js, Next.js, MongoDB, auth0 for user authentication and REST API
- Created natural language processing applications using LLM's and Python for sentiment analysis and chatbots
- Built and trained a reinforcement neural network using TensorFlow, KerasRL for predicting stock prices
- Developed Serverless REST API with C # and Azure for IoT Sensors and data analytics using ASP.NET
- Developed low level ARM embedded systems with C whilst using memory allocation and register programming
- Engineered a high performance C++ linear algebra library for data science through Jenkins, Docker and Cmake

IT administrator

Mesopotamia Group

📅 Jun 2020 - Mar 2021 (10 Months)

📍 Amman Jordan

- Collaborated with a team to build a local file server on Ubuntu built using Apache and Nextcloud
- Administered the server using Bash, Python and Nmap with SSH tunneling and port forwarding
- Engineered a load balancer using MPI and C to distribute traffic between servers increasing availability by 34%
- Analyzed the network data logs using Python to predict and prevent cyber attacks
- Implemented Linux networking and protocols using IP addressing and routing for cloud hosting

📖 Education

BEng (Hons) in Electronics and Computer

University of Leeds (UoL)

📅 Sept 2019 - June 2024

✅ First-Class Honors (80 %)

- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design
- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Research

Skills and Languages

Technical: C \ C++ Python Java C# JavaScript CI \ CD DevOps Bash Linux Docker
AWS Jira Git Pulumi Kubernetes MySQL MongoDB PostgreSQL .NET WPF React Js
Flutter Flask Express.js Springboot NumPy KerasRL Pytorch TensorFlow Pandas OpenCv
Node.js OpenGL OpenMP CUDA Arduino Blender Photoshop AutoCAD

Soft: Management Leadership Motivated Deterministic Communication Problem solver

Languages: Arabic (Full Professional) English (Full Professional) Spanish (Elementary)

Publications

Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development  2024

Pending publication - 1st Author

- Developed a full stack chat bot web application with Typescript, React.js, Next.js, TailwindCSS and MongoDB
- Programmed language processing algorithms using large language models to interpret and execute messages
- Developed a Python Flask REST Server with SQLite on the Raspberry Pi to communicate with the chat bot
- Engineered a circuit with sensors, fans, LCD and buttons to execute the user's commands from the chat bot

Projects

MCPP: High Performance C++ Library for Numerical Computation  Apr 2021 - Oct 2022  Personal

- Created a cross-platform math library for C++ used for matrix algebra, statistics and machine learning.
- Engineered with CI/CD using Jenkins, unit tests, Docker, Cmake, Doxygen and Git for version control
- Improved performance by 40% using parallel computing with OpenMP, OpenCL and CUDA

TBot: Machine Learning Stocks Trading Bot  Apr 2021 - Oct 2021  Personal

- Developed gradient boosting machine learning model using Python and Scikit-learn for stock prediction
- Analyzed and processed data using Pandas, Numpy and Scipy for training and testing the model
- Backtested the model using historical data and simulated trading using Python and Matplotlib
- Developed a REST API and a user interface to interact with the bot using AWS lambda, Flask and PyQt

Professional Courses

Data Analysis with Python  Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory data analysis and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate regression models using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python  Sept 2021

- Analyze data within a relational database on the cloud using IBM DB2, SQL, Python
- Construct SQL CRUD operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development  Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, Numpy, and Beautiful Soup
- Access web data using APIs and web scraping from Python in Jupyter Notebooks.

Awards

2nd highest mark on entire cohort  Jul 2024

University of Leeds

 Leeds, UK

Future leaders top 3  May 2023

Johnson Controls

 London, UK

Best use of auth0  Feb 2023

Royal Hackaway v6

 London, UK